

# Linear Algebra I: HW2 Partial Solution

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1.4-1. TFTFTF

1.4-5. YNNYYNYN

1.4-13. Given  $s \in \text{span}(S_1)$ , there exists  $v_1, \dots, v_n \in S_1$  and  $a_1, \dots, a_n \in F$  such that  $s = a_1v_1 + \dots + a_nv_n$ , but  $v_1, \dots, v_n \in S_1 \subseteq S_2$ , so  $s \in \text{span}(S_2)$ . If we further assume that  $\text{span}(S_1) = V$ , then  $V = \text{span}(S_1) \subseteq \text{span}(S_2) \subseteq V$ , which forces that  $\text{span}(S_2) = V$ .

1.4-14. Since  $S_1, S_2 \subseteq S_1 \cup S_2$ , we have  $\text{span}(S_1), \text{span}(S_2) \subseteq \text{span}(S_1 \cup S_2)$ , then by the minimality of sums,  $\text{span}(S_1) + \text{span}(S_2) \subseteq \text{span}(S_1 \cup S_2)$ . Conversely, since  $S_1 \cup S_2 \subseteq \text{span}(S_1) + \text{span}(S_2)$ , by the minimality of [spans](#),  $\text{span}(S_1 \cup S_2) \subseteq \text{span}(S_1) + \text{span}(S_2)$ .

1.5-1. FTFFTT

1.5-2. DIIDDIDIID

1.5-15. ( $\implies$ ) Suppose  $S$  is linearly dependent, if  $u_1 = 0$ , then we are done. Otherwise,  $\{u_1\}$  is linearly independent. If  $\{u_1, u_2\}$  is linearly dependent, then  $u_2 \in \text{span}\{u_1\}$ , and we are done (here  $k = 1$ ). Otherwise,  $\{u_1, u_2\}$  is linear independent. By doing so repeatedly, we can find a  $k$ ,  $1 \leq k < n$  such that  $\{u_1, \dots, u_k\}$  is linearly independent and  $\{u_1, \dots, u_{k+1}\}$  is linearly dependent. Consequently,  $u_{k+1} \in \text{span}\{u_1, \dots, u_k\}$ . ( $\impliedby$ ) If  $u_1 = 0$ , then  $0 \in S$  so  $S$  is linear dependent. If  $u_{k+1} \in \text{span}\{u_1, \dots, u_k\}$  for some  $k$ , then there is a relation

$$u_{k+1} = a_1u_1 + a_2u_2 + \dots + a_ku_k, \text{ for some } a_1, \dots, a_k \in F \text{ not all zero}$$

Thus there is a nontrivial relation

$$a_1u_1 + a_2u_2 + \dots + a_ku_k - u_{k+1} = 0,$$

so  $S$  is linearly dependent.

1.5.20 Suppose on the contrary that there is a nontrivial relation

$$a_1e^{rt} + a_2e^{st} = 0.$$

Plug in  $t = 0$ , we have  $a_1 + a_2 = 0$ , thus  $a_1 = -a_2$  and  $a_1, a_2 \neq 0$ . Plug in  $t = 1$ , we have  $a_1(e^r - e^s) = 0$ , so  $e^r = e^s$ , which implies  $r = s$  and contradicts our assumption  $r \neq s$ .

3. The proof is mostly the same as that of 1.4.14. One needs to keep in mind that the sum of a family of vector subspaces collects vectors which are *finite sums* of vectors from the family.